

Hammad Ahmad

AI/ML Researcher — Trustworthy NLP · Retrieval-Augmented Generation · Knowledge Graphs

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#) | [ORCID](#)

RESEARCH PROFILE

AI/ML researcher with an MSc (Merit) in Applied AI and a first-author peer-reviewed publication. My work centres on reliable, transparent language systems — retrieval-augmented generation grounded in knowledge graphs — alongside interpretable predictive modeling on high-dimensional real-world data. I am seeking a PhD to develop this into rigorous research on trustworthy, knowledge-grounded NLP.

EDUCATION

University of Bradford Sep 2024 – Sep 2025

MSc, Applied Artificial Intelligence and Data Analytics — Merit Bradford, UK

- **Dissertation:** FinLaw-UK — A Graph-Augmented Retrieval Chatbot for Reliable UK Financial Regulation.

COMSATS University Islamabad Sep 2020 – Jul 2024

BS, Bioinformatics Islamabad, Pakistan

- **Thesis:** AI-Assisted Modeling and Risk Prediction for High-Dimensional Clinical Data.

PUBLICATIONS

Ahmad, H., Khan, M.U., & Azam, M. (2024). Comparative Analysis of Machine Learning Methods for Enhancing Sleep Efficiency and Prediction. *ICSMAI 2024* (Springer). DOI: 10.1007/978-3-031-66854-8_1

RESEARCH EXPERIENCE

University of Bradford Jan 2025 – Sep 2025

Research Assistant — Machine Learning & LLMs Bradford, UK

- Designed FinLaw-UK, a retrieval-augmented generation system pairing an open-source LLM (Mistral 7B) with a Neo4j knowledge graph to improve factual reliability in UK financial-regulation question answering.
- Fine-tuned Sentence Transformer embeddings and introduced cross-encoder re-ranking to raise retrieval precision.
- Evaluated answer faithfulness and relevance with the RAGAS framework, establishing a reproducible benchmarking pipeline.

COMSATS University Islamabad Jul 2023 – Jul 2024

Research Assistant — Data Science Islamabad, Pakistan

- Developed ensemble predictive models (Random Forest, Gradient Boosting) on high-dimensional clinical data, reaching ~93% classification accuracy under stratified cross-validation.
- Applied SHAP-based interpretability to identify influential predictors and improve model transparency.

INDUSTRY EXPERIENCE

Outlyst Oct 2025 – Mar 2026

AI / Machine Learning Engineer Leeds, UK (Remote)

- Developed backend dialogue-flow logic and workflow automation for an AI voice-agent system (Retell AI, FastAPI), integrating LLM-based dialogue with production infrastructure.

TECHNICAL SKILLS

Machine Learning & NLP : LLMs, Retrieval-Augmented Generation (RAG), Sentence Transformers, cross-encoder re-ranking, RAGAS evaluation, predictive modeling, ensemble methods, SHAP, PyTorch, TensorFlow, scikit-learn

Knowledge & Data : Neo4j knowledge graphs, vector embeddings, semantic search, PostgreSQL, pandas, NumPy

Programming & Tools : Python, C++, SQL, FastAPI, Docker, Git, Linux

LANGUAGES

English (IELTS 7.0, CEFR C1) | Urdu (Native) | German (A1.2)